

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	08
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Simulation and Deployment of Fog Computing Architecture for Smart IoT Applications.
Theory (100 words)	Fog Computing is an extension of cloud computing designed to bring computation, storage, and networking services closer to IoT devices. In a traditional cloud-based system, data collected by sensors is transferred to a remote cloud server for processing. This can cause higher network traffic, increased delay, and dependence on internet connectivity. Fog Computing addresses these limitations by introducing intermediate computing nodes between IoT devices and the cloud. In a fog-based system, IoT devices continuously generate data from sensors and send it to a nearby fog node. The fog node performs operations such as data filtering, aggregation, analysis, and local decision-making. For example, if a temperature sensor detects an abnormal temperature, the fog node can immediately generate an alert without waiting for the cloud to process the information. Important or summarized data can then be transmitted to the cloud for permanent storage and advanced analysis. The architecture of Fog Computing can be represented as:

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	Sensors/IoT Devices → Fog Node → Cloud Server The major characteristics of Fog Computing include low latency, location awareness, distributed processing, reduced bandwidth consumption, and real-time response. Since processing takes place closer to the source of data, fog computing is suitable for applications where quick decisions are required. Fog Computing is commonly used in smart healthcare, smart homes, smart cities, industrial monitoring, connected vehicles, traffic control, surveillance systems, and environmental monitoring. It provides a bridge between resource-constrained IoT devices and powerful cloud infrastructure. In this practical, a computer can act as a simulated fog node. Sensor or IoT data is received by the computer, processed locally according to predefined conditions, and the required information is forwarded to a cloud platform. This demonstrates the basic concept of distributed processing and shows how fog computing can improve the efficiency of IoT applications.
Procedure and Execution (100 Words)	1. Create a folder named FogIoT in the C: drive. 2. Inside the folder, create a Python program named <code>iot_device.py</code> to simulate an IoT device. 3. Write the program to generate temperature and humidity sensor readings. 4. Execute <code>iot_device.py</code> and check whether the sensor values are generated correctly. 5. Create a second Python program named <code>fog_node.py</code> to simulate the Fog Node. 6. Configure the Fog Node to receive temperature and humidity data from the simulated IoT device. 7. Implement local processing rules in <code>fog_node.py</code> to classify the temperature as Normal or Abnormal. 8. Run the Fog Node and transmit the generated IoT sensor readings to it. 9. Allow the Fog Node to process the received data locally and display the temperature, humidity, and processing result. 10. Test the system using different temperature values to verify the local decision-making of the Fog Node. 11. Record the IoT sensor input and corresponding Fog Node output as observations. 12. Verify that the sensor data is processed at the Fog Layer before being considered for further processing or storage at the Cloud Layer.



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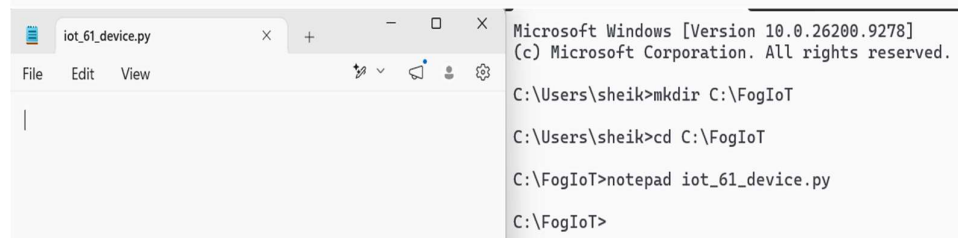
Stepwise Screenshots with steps:

```
Microsoft Windows [Version 10.0.26200.9278]  
(c) Microsoft Corporation. All rights reserved
```

```
C:\Users\sheik>mkdir C:\FogIoT
```

```
C:\Users\sheik>cd C:\FogIoT
```

```
C:\FogIoT>|
```



```
C:\FogIoT>python iot_61_device.py  
IoT Device Started
```

```
-----  
sheikh wasimuddin lab HCS  
Temperature: 21.77 °C  
Humidity: 43.83 %  
-----
```

```
sheikh wasimuddin lab HCS  
Temperature: 30.2 °C  
Humidity: 64.98 %  
-----
```

```
sheikh wasimuddin lab HCS  
Temperature: 35.46 °C  
Humidity: 59.54 %  
-----
```



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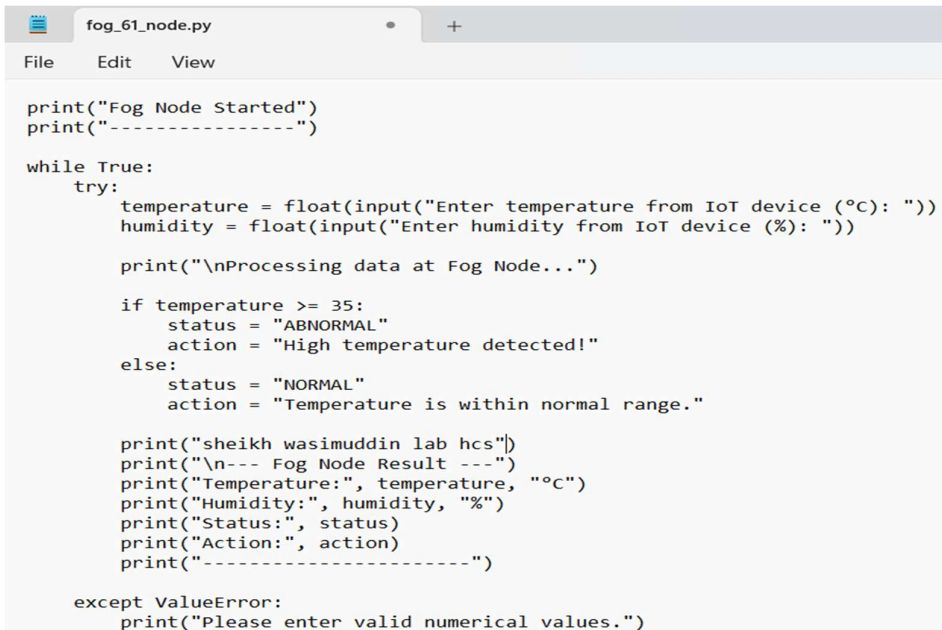
To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

```
import random
import time

print("IoT Device Started")
print("-----")

for i in range(10):
    temperature = round(random.uniform(20, 45), 2)
    humidity = round(random.uniform(40, 80), 2)
    print("sheikh wasimuddin lab HCS")
    print("Temperature:", temperature, "°C")
    print("Humidity:", humidity, "%")
    print("-----")

    time.sleep(2)|
```



```
fog_61_node.py
File Edit View

print("Fog Node Started")
print("-----")

while True:
    try:
        temperature = float(input("Enter temperature from IoT device (°C): "))
        humidity = float(input("Enter humidity from IoT device (%): "))

        print("\nProcessing data at Fog Node...")

        if temperature >= 35:
            status = "ABNORMAL"
            action = "High temperature detected!"
        else:
            status = "NORMAL"
            action = "Temperature is within normal range."

        print("sheikh wasimuddin lab hcs")
        print("\n--- Fog Node Result ---")
        print("Temperature:", temperature, "°C")
        print("Humidity:", humidity, "%")
        print("Status:", status)
        print("Action:", action)
        print("-----")

    except ValueError:
        print("Please enter valid numerical values.")
```



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	<pre>C:\FogIoT>notepad fog_61_node.py C:\FogIoT>python fog_61_node.py Fog Node Started ----- Enter temperature from IoT device (°C): 36 Enter humidity from IoT device (%): 28 Processing data at Fog Node... sheikh wasimuddin lab hcs --- Fog Node Result --- Temperature: 36.0 °C Humidity: 28.0 % Status: ABNORMAL Action: High temperature detected! -----</pre>
Conclusion	The Fog Computing practical was successfully implemented using Python by simulating an IoT device and a Fog Node. The IoT device generated temperature and humidity readings, which were transmitted to the Fog Node for local processing. The Fog Node analyzed the received values and identified normal and abnormal temperature conditions. This experiment demonstrated the role of Fog Computing in reducing processing delays and enabling quick decisions near the data source before forwarding information to the Cloud layer.
Date	02/10/2026